# DIP BHATTACHARJEE

**Computer Science & Engineering Undergraduate | AI/ML Research Enthusiast**

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## PROFESSIONAL SUMMARY

Computer Science and Engineering undergraduate majoring in **Computational Theory**, with a strong interest in Artificial Intelligence, Machine Learning, Natural Language Processing, Deep Learning, and applied research. Academic training includes Machine Learning, Natural Language Processing, Graph Theory, Data Structures and Algorithms, and Object-Oriented Programming. Experienced in programming, software development, AI/ML-based projects, and solving computational problems. Passionate about developing practical, data-driven solutions and continuously expanding technical expertise in AI/ML, NLP, and emerging intelligent technologies.

## EDUCATION

**Bachelor of Science in Computer Science and Engineering** — **Aug 2023 – Present**
*American International University-Bangladesh* — *CGPA: 3.83/4.00*

**Major:** Computational Theory

**Completed Major Courses:** Natural Language Processing, Machine Learning, Basic Graph Theory.

**Additional Relevant Coursework:** Data Structures and Algorithms, Object-Oriented Programming with Java, Programming in C++, and Discrete Mathematics.

**Higher Secondary Certificate (HSC)** — **2020 – 2022**
*Mohipal Government College – Science* — *GPA: 4.89/5.00*

**Secondary School Certificate (SSC)** — **2018 – 2020**
*Batisa Secondary School – Science* — *GPA: 4.89/5.00*

## RESEARCH & CONFERENCE PAPERS

**Progressive Two-Stage Transfer Learning with Dual Imbalance Compensation for Interpretable Enamel Caries Classification**
*Accepted for Presentation at the 5th IEEE International Conference on Signal Processing, Information, Communication and Systems (SPICSCON 2026)*

**Development of a Cloud-Linked Low-Cost Machine Learning-Enabled Weather Monitoring System**
*Published in the Proceedings of IEEE TENSYMP 2026 · DOI: 10.1109/TENSYMP68909.2026.11704037*

**Diagnostic-Guided Monte Carlo Risk Estimation Using GARCH, EGARCH, and Bootstrap Models**
*Accepted for Presentation at the 3rd IEEE International Conference on Computing, Applications and Systems (COMPAS 2026)*

## TECHNICAL SKILLS

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| **Programming:** | Python, C, C++, Java, C#, JavaScript |
| **AI & Machine Learning:** | Machine Learning, Deep Learning, Transfer Learning, Classification, Model Training & Evaluation |
| **Natural Language Processing:** | Text Preprocessing, Text Classification, Feature Extraction, NLP Fundamentals |
| **Data & ML:** | Data Preprocessing, Data Analysis, Model Evaluation, Class Imbalance Handling |
| **Core CS:** | Data Structures & Algorithms, Graph Theory, Object-Oriented Programming, Problem Solving |
| **Research Areas:** | Computer Vision, Medical Image Analysis, Explainable AI, NLP, IoT-based Intelligent Systems |
| **Tools:** | Microsoft Office, AutoCAD, Adobe Photoshop |

## PROJECTS

**Hospital Management System** *Java*

- Developed a system to support essential hospital operations, including patient records, appointments, billing, and staff management.
- Applied object-oriented programming principles in Java to design structured and maintainable system functionality.

**Poultry Farm Management System** *Java, Database*

- Developed a database-driven management system to organize and manage poultry farm-related information efficiently.
- Applied Java programming and database concepts to implement structured data storage, management, and system operations.

## EXPERIENCE

**Instructor** *NextGen Coaching Center for Admission Preparation*

- Taught key academic subjects required for university admission preparation, helping students strengthen fundamental concepts.
- Developed mock tests and quizzes to improve students' problem-solving speed, accuracy, and examination preparation.
- Conducted one-on-one mentoring sessions to help students understand challenging topics and improve their academic confidence.

**Freelance Tutor** *Programming & Problem Solving*

- Mentored students in Python, JavaScript, and C++, strengthening programming fundamentals and problem-solving skills.
- Designed personalized learning plans covering programming, data structures, algorithms, and computational problem solving.
- Guided students through practical coding exercises and project-based learning activities.

## ACHIEVEMENTS & CERTIFICATIONS

- **Dean's Award**

## ADDITIONAL INFORMATION

**Languages:** Bangla (Native), English (Fluent), Hindi (Proficient in Listening, Basic Speaking)
**Research Interests:** Artificial Intelligence, Machine Learning, Natural Language Processing, Deep Learning, Computer Vision, Explainable AI, Medical Image Analysis, and Intelligent Systems